

KYUNGMIN KIM

kyungminkim@snu.ac.kr · kyungmax.github.io

RESEARCH INTERESTS

Database Systems, with focus on Vector Search and Cloud-native Architectures

EDUCATION

Seoul National University (SNU)

Mar. 2025 - Present

M.S. in Data Science

GPA: 4.05 / 4.3

Sungkyunkwan University (SKKU)

Mar. 2016 - Feb. 2023

B.S. in Computer Science & Engineering

*Including 18 months of Korean military service

GPA: 4.05 / 4.5

Magna Cum Laude

RESEARCH & PROFESSIONAL EXPERIENCE

Database Systems Research Laboratory, Seoul National University

Mar. 2025 - Present

Graduate Research Assistant (Advisor: Prof. Sang-Won Lee)

Seoul, South Korea

- Designed an adaptive search method for HNSW-based vector databases that infers per-query difficulty from internal search signals and adjusts search effort at runtime, improving query throughput at matched recall. [1]
- Researched materialized views on data lakehouses (Apache Iceberg, Apache Spark), addressing view maintenance and query rewriting over open table formats. [2]

Kakao Corp.

Dec. 2021 - Apr. 2025

Data Engineer, Advertisement (AD) Development Team

Seongnam, South Korea

*Kakao offers KakaoTalk, the largest messenger application in South Korea with 45M+ users.

- Built and operated real-time data pipelines processing advertising traffic up to 100K events/s with Apache Kafka and Flink on Hadoop and Kubernetes clusters.
- Migrated 40+ production Flink jobs from YARN to Kubernetes with autoscaling, reducing resource usage and operational cost while improving resilience to traffic spikes.
- Designed storage schemas and selected backends across Elasticsearch and Cassandra for low-latency querying of high-volume event logs.

Kakao Corp.

Summer, 2021

Software Engineer Intern, AD Development Team

Seongnam, South Korea

- Designed and implemented a prototype for a display AD system, including a demand-side platform, AD serving, billing, and reporting components.
- Developed low-latency web servers with Java and Spring Boot and orchestrated data pipelines on Kubernetes with Kafka and ELK stacks for logging and analytics.

Robotics Research Laboratory, Purdue University

Aug. 2018 - Dec. 2018

Undergraduate Research Assistant (Advisor: Prof. Eric T. Matson)

West Lafayette, IN, United States

- Built a UAV-based thermal-imaging pipeline to detect defective solar-panel cells, using OpenCV for contour-based segmentation and shape matching.
- Quantified how panel tilt angle affects detection accuracy, identifying the angular range over which thermal fault detection stays reliable. [5]

Database Systems Research Laboratory, Sungkyunkwan University

Mar. 2018 - Aug. 2019

Undergraduate Research Assistant (Advisor: Prof. Sang-Won Lee)

Suwon, South Korea

- Analyzed and optimized database engine performance on non-volatile memory storage. [3]
- Researched MySQL/InnoDB MVCC architecture and locking schemes to identify opportunities for enhancing database performance. [4]

PUBLICATIONS

- [1] “Start Wide, Cut Early: Self-Calibrating Adaptive HNSW Search”
Kyungmin Kim, Dongseob Kim, Jihyo Jang, Joobo Shim, Jaeyoung Do, Sang-Won Lee
Submitted to *Proceedings of the VLDB Endowment (PVLDB)*, Aug. 2026, **Under Review**
- [2] “Supporting Materialized Views on Data Lakehouses” **[Best Paper Award]**
Kyungmin Kim, Sang-Won Lee
Korean DataBase Conference (KDBC 2025), Nov. 2025
- [3] “Performance evaluation on concurrency control techniques of main-memory DBMS based on log device”
Kyungmin Kim, Jonghyeok Park, Sang-Won Lee
Korean DataBase Conference (KDBC 2022), Nov. 2022
- [4] “Analysis of MySQL/InnoDB Lock Contention on TPC-C Workload”
Kyungmin Kim, Jong-Hyeok Park, Mijin An, Sang-Won Lee
Korea Computer Congress (KCC 2019), Jul. 2019
- [5] “Analyzing the range of angles of a solar panel to detect defective cells, using a UAV”
Kyungmin Kim, Yoojin Choi, Keonyoung Shim, Hyewon Jeon, Jane Commerford, Eric T. Matson
Third IEEE International Conference on Robotic Computing (IRC 2019), Feb. 2019

HONORS & AWARDS

Best Paper Award , Korean DataBase Conference (KDBC)	Nov. 2025
Graduate Student Research Grant , National Research Foundation of Korea (KRW 12M)	Sep. 2025 - Aug. 2026
SKKU Magna Cum Laude , SKKU	Feb. 2023
Full Tuition Scholarship , SKKU (approx. KRW 36M)	Spring 2016 - Fall 2022
Creativity Award in 2020 Army Entrepreneurship Contest , Republic of Korea Army	Jun. 2020
Excellence Paper Award , Korea Computer Congress (KCC)	Aug. 2019
K-SW Purdue Scholarship , South Korean Government (IITP)	Aug. 2018
Dean’s List , SKKU	Spring 2016; Spring 2017; Spring 2019; Fall 2021

TEACHING EXPERIENCE

Seoul National University	Spring 2026
Teaching Assistant	
· M3674.000700: Big Data and Knowledge Management Systems for Data Science Applications	

PATENTS

Kyungmin Kim, Hyeonchul Yoon. “Method for Searching Music Using Humming Analysis. KR-Patent 10-2018-0046898

PROFESSIONAL SERVICE

Session Chair Database Systems for Advanced Applications (DASFAA)	Apr. 2026
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TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL
Systems: Apache Flink, Apache Kafka, Apache Hadoop, Apache Spark, Kubernetes
Databases: MySQL, Elasticsearch, Apache Cassandra, Apache Iceberg, Redis